

Data Analysis on Vehicle CO2 Emissions

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Introduction

As climate change increasingly threatens the environment, the need to better understand and reduce carbon dioxide (CO₂) emissions has grown among environmentally conscious stakeholders. Among greenhouse gas sources, transportation remains one of the largest contributors with personal and commercial vehicles playing a substantial role. According to the Canadian government, transportation accounted for 23% of Canadian greenhouse gas emissions in 2023 (Government of Canada). Because of the large proportion of emissions Canadian transportation vehicles are responsible for, our team was interested in using data to better understand the relationship between vehicle characteristics and the amount of CO₂ - a proxy for greenhouse gases - they emit.

Because greenhouse gas emissions have steadily increased over time, we saw the potential of using historical vehicle data to analyze fuel consumption and ultimately build a model to predict CO₂ emissions. Since several factors, including vehicle fuel type and consumption rate, affect emission levels our guiding question for this project was to identify the most influential factors on vehicle emissions and build a model that can predict vehicle CO₂ emissions. This kind of predictive capability could be useful for a variety of stakeholders including consumers and policymakers. Equipped with the ability to better understand the environmental impact of a given vehicle, environmentally conscious consumers can feel empowered to make purchases that align with their values and minimize their impact on climate change. Similarly, policymakers can use these findings to support arguments for more effective emissions regulations.

Dataset Description

The dataset used in this project is the CO₂ Emission by Vehicles dataset by Debajyoti Podder from Kaggle which compiles data originally sourced from the Government of Canada's official open data platform. We chose to use this dataset because it compiles data relevant to different vehicle types in one dataset, reflecting the diverse landscape of the vehicle market. The dataset contains data regarding vehicle emissions spanning a 7-year time period, including 7,385 entries and 12 attributes related to various vehicle specifications and performance metrics. Each record represents a unique vehicle model and includes the following columns:

- Make – Manufacturer or brand of the vehicle (e.g., Acura, Ford, Toyota).
- Model – Specific vehicle model name (e.g., AWD, Sedan, Convertible)
- Vehicle Class – Category of the vehicle (e.g., COMPACT, SUV, PICKUP TRUCK).
- Engine Size (L) – Engine displacement in liters.
- Cylinders – Number of engine cylinders.
- Transmission – Type of transmission and number of gears including A6 (Automatic, 6-speed), M5 (Manual, 5-speed), AV (Continuously variable)
- Fuel Type – Type of fuel used including X or regular gasoline, Z or premium gasoline, d or diesel, e or ethanol (E85), and N or natural gas.

- Fuel Consumption City (L/100 km) – Fuel usage measured in liters per 100 km in city driving conditions.
- Fuel Consumption Hwy (L/100 km) – Fuel usage in highway conditions.
Fuel Consumption Comb (L/100 km) – Combined fuel consumption (weighted average of the previous two variables with the following weights: 55% city, 45% highway).
- Fuel Consumption Comb (mpg) – Combined fuel consumption in miles per gallon.
- CO2 Emissions (g/km) – **The primary target outcome variable:** grams of CO2 emitted per kilometer driven (combined city and highway).

This dataset allows for a detailed analysis of how various technical and design features impact a vehicle's carbon footprint. Because it reflects real-world testing across multiple years and includes a wide range of vehicle types, it serves as a complete foundation for building predictive models and conducting environmental impact assessments.

Dataset Cleaning and Prep Process

To ensure the dataset was suitable for analysis, we began by cleaning and preparing the data. We first removed any rows containing missing values to maintain the integrity of our results. Next, we sorted the dataset for consistency and reset the index allowing us to see records with similar vehicle characteristics (make, model, vehicle class) and eliminate duplicates. Through this cleaning process we ended up identifying and removing 1,103 duplicate records, eliminating redundancy in our data and ensuring each vehicle configuration appeared only once.

The next step in our cleaning process was preparing the data for regression analysis. To do this we encoded categorical variables like model or vehicle class into a numerical format using one-hot encoding ensuring all predictor variables were numerical.

For our regression analysis we defined all relevant vehicle features as predictors and CO2 emissions (in grams per kilometer) as the target outcome variable. Before proceeding with the analysis we also added a constant term to the predictor variables to ensure that the model could produce an intercept, accounting for baseline cases where all predictor variables equal 0. After verifying that all features were in the correct numerical format, we proceeded with statistical modeling and feature selection using ordinary least squares regression and Lasso.

Data Analysis Technique 1: Testing Assumptions of Linear Regression

Before conclusions can be drawn about our linear regression model, we must test the assumptions of linear regression. These assumptions include linearity in parameters, constant variance, mutually independent residuals, and normally distributed residuals.

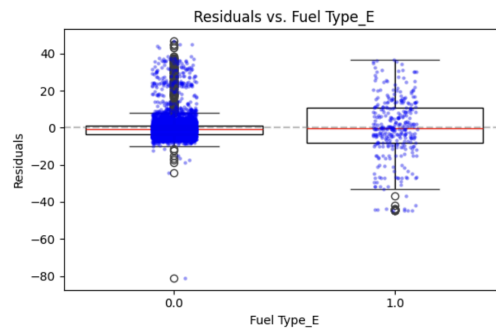
Linearity in parameters

First, we checked linearity by plotting the residuals against the predictors, some of which are visualized below. Since our predictors include both categorical and continuous variables, we analyzed them separately:

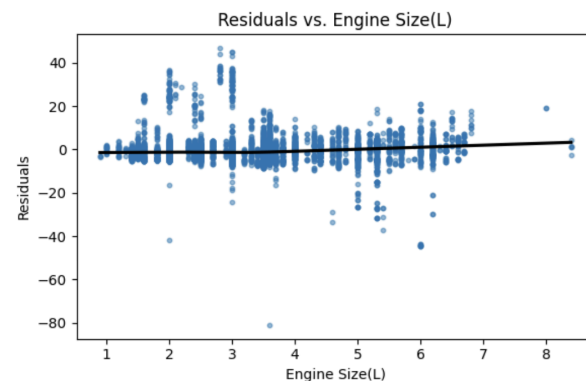
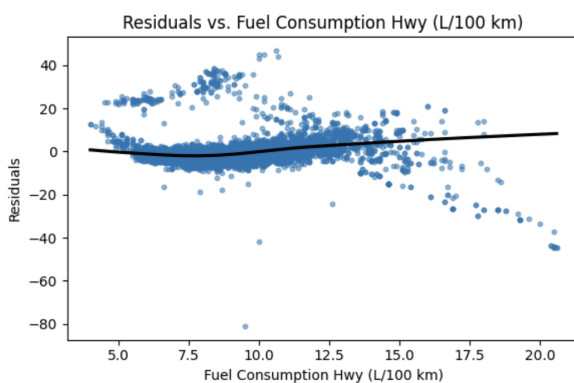
1. For continuous predictors (e.g., Fuel Consumption Hwy (L/100 km)) we plotted a LOWESS line over the residuals.
2. For categorical predictors (e.g., Fuel Type), we used box plots of residuals per category to check if the expected residuals $E[\epsilon]$ were approximately zero.

The residual plots indicate that the linearity assumption holds for most of our predictors. For example:

1. In the Residuals vs. Ethanol Fuel Type box plots, the expected residuals, denoted by the red line, is close to 0 for both plots, indicating that linearity holds for this covariate.



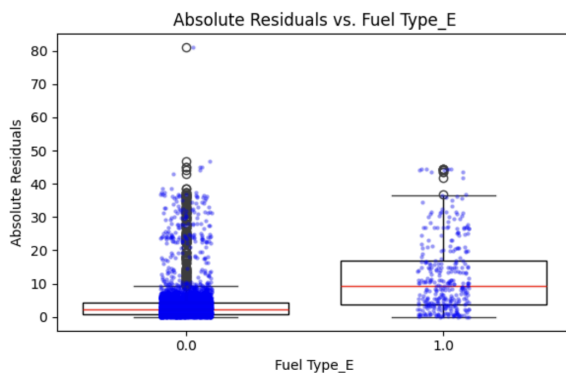
2. In our scatterplots with the LOWESS curve, the LOWESS curve is approximately centered at the zero residual and it is roughly flat, showing that linearity holds for the continuous predictors.



Constant Variance

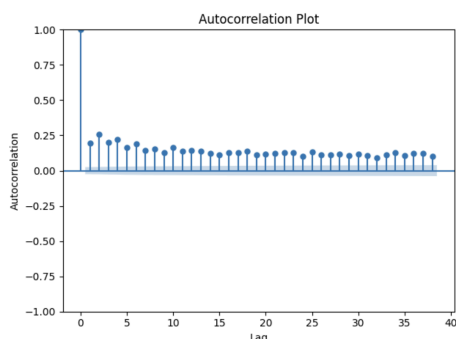
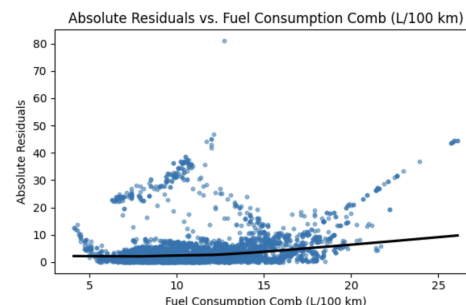
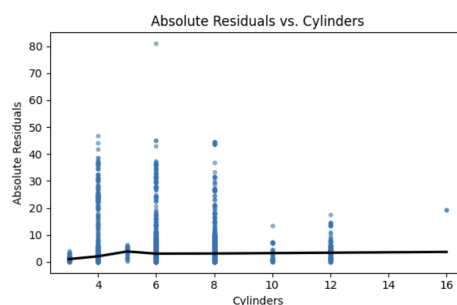
We then tested for constant variance, again analyzing categorical and continuous variables separately (again, we'll only display a few here):

1. For continuous predictors, we plotted a LOWESS line for the absolute residuals.
2. For categorical predictors, we plotted a box plot for each group and checked if they had similar spread (box height).



Based on these results we can deduce that overall, constant variance does not hold for this model. This is because in the box plots for categorical predictors, there is a larger spread in residuals for one fuel type compared to the other as observed through the difference in box heights.

As for the continuous variables, although noise could be a factor explaining the increasing slope on some of our plots' LOWESS curves, as a whole our model has heteroskedasticity and this assumption does not hold. To combat this, we tried to improve our model's constant variance by experimenting with log transforming, square rooting, as well as taking the reciprocal of the dependent variable, however, in all cases heteroskedasticity remained in our model. This remains a valid limitation that will be discussed later in the report.

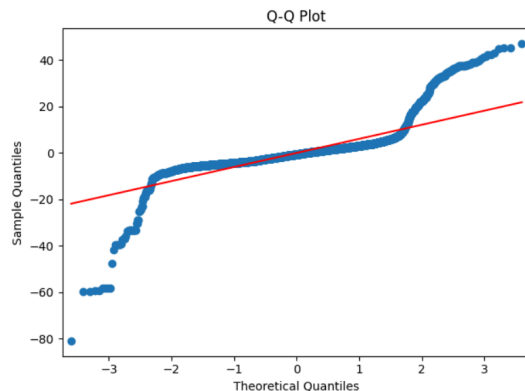


Mutually Independent Residuals

Next, we tested for mutually independent residuals by using an autocorrelation plot. Since all of the autocorrelation values are well above the highlighted blue range, this indicates that there

is a positive autocorrelation and the assumption does not hold. We attempted to improve our independence by introducing lagged variables to our model, however the lack of a time component in the dataset made this unsuccessful; this is a limitation that will also be discussed later in the report.

Normally Distributed Residuals



Finally, we checked normality by using a Q-Q plot. The plot shows a convex-concave pattern, which means that the distribution of the residuals is light-tailed. A solution to this issue would be to check for collinearity and remove predictors that have high correlations between them.

Collinearity

We began by assessing multicollinearity using Variance Inflation Factor (VIF), which quantifies how much a predictor's variance is inflated due to its correlation with other predictors. A VIF near 1 indicates little to no collinearity, while a VIF above 10 suggests serious multicollinearity that can distort coefficient estimates and make model interpretation unreliable. The initial results (shown in the appendix) revealed extremely high VIF values for Fuel Consumption City, Fuel Consumption Hwy, and Fuel Consumption Comb, all far exceeding the threshold of 10, with the combined consumption variable reaching over 5000. These values indicate that these predictors are highly correlated with each other, making it difficult to isolate their individual effects on the response variable. Since multicollinearity affects inference rather than prediction, it was necessary to remove redundant variables to improve interpretability. After removing the collinear variables, all remaining predictors had VIFs well below 10, with most close to 1, indicating a substantial reduction in collinearity. The final model includes only Engine Size, Cylinders, Fuel Consumption Comb (L/100 km), and Fuel Type_E, all of which contribute uniquely to the model without introducing multicollinearity. Thus, we retained these variables for the final model and did not require further transformations such as orthogonalization.

Data Analysis Technique 2: Lasso Regression and Linear Regression

To predict CO2 emissions (g/km), we fitted a linear regression model leveraging Lasso regression for feature selection to improve model performance and interpretability. Lasso is particularly useful when dealing with datasets that include multiple predictors because it applies an L1 penalty that shrinks less informative coefficients toward zero. This gives us a sparser model with fewer parameters. By reducing model complexity, Lasso also helps prevent overfitting.

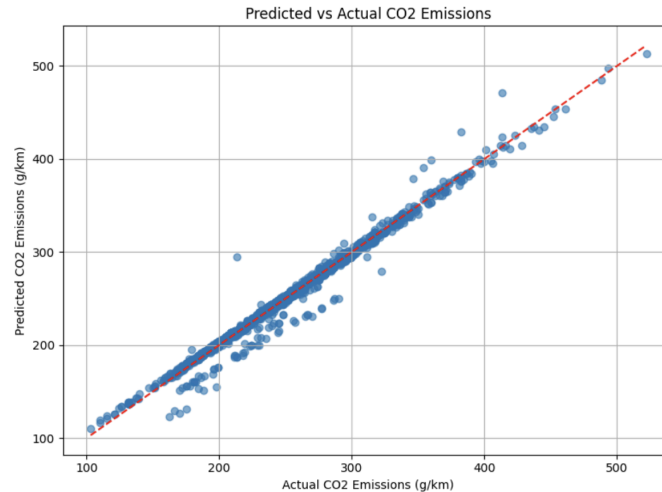
The selected features via Lasso were engine size, number of cylinders, combined fuel consumption, and fuel type. We then fit a final OLS model using these features. The adjusted R-squared value of 0.985 indicates that about 98.5% of the variation in CO2 emissions is explained by our model, showing a very strong fit. The F-statistic is extremely high (1.019×10^5) with a p-value of 0.000, confirming the overall significance of the model. Among the predictors, fuel consumption has the largest positive effect on emissions (coef = 21.7851), and is highly significant ($p < 0.001$). Fuel type (E) shows a strong negative relationship (coef = -111.02), suggesting that vehicles using this fuel type emit significantly less CO2, and is also statistically significant. The number of cylinders also has a positive and significant effect. Engine size has a positive but borderline insignificant effect ($p = 0.094$), indicating weaker evidence of its impact. Overall, the regression model is statistically sound and explains the outcome variable well, but it's important to check assumptions and consider robustness before making firm conclusions.

As explained in the earlier section, some linear regression assumptions were violated and cannot be corrected. As a result, the results from the model may be unreliable. Since our goal is to predict future outcomes, we leveraged train/test splits to assess the model's predictive ability despite these violations.

Data Analysis Technique 3: Leveraging Train/Test Splits

To evaluate the model's performance on unseen data, we implemented a train/test split approach. This involves dividing the dataset into two parts: one to train the model and the other to test how well it generalizes. After training the linear regression model on the training set, we evaluated its predictive accuracy on the test set. The resulting Mean Squared Error (MSE) was 65.48, and the Average Squared Error was 4.34 - both indicating a relatively low average difference between the predicted and actual CO2 emissions.

Furthermore, the correlation between the predicted and actual values was 0.9910, suggesting that the model captures the underlying trends in the data very effectively. A scatter plot of predicted vs. actual values (seen below) confirms this: the points fall closely along the diagonal line, visually demonstrating the model's accuracy. This strong alignment indicates that the model not only fits the training data well but also performs robustly on new, unseen data, reinforcing its reliability.



Conclusion & Limitations

Our report sought to investigate the key factors that influence vehicle CO₂ emissions as well as develop a predictive model that estimates emissions using vehicle characteristics. To achieve this, we first cleaned our data and used Lasso regression to select our features, which were engine size, number of cylinders, combined fuel consumption, and fuel type. We checked the assumptions for linear regression and fit an OLS model using the selected features, with our model indicating that our covariates explain the outcome variable fairly well. Finally, we used training and testing sets to demonstrate our model's good predictive accuracy and validate its applicability in real-world scenarios. Specifically, real-world stakeholders like consumers and policymakers can use this model as a tool to predict vehicle CO₂ emissions allowing them to conceptualize the environmental impact of such vehicles before they are purchased or produced. Thus, resources don't need to be diverted towards creating vehicle prototypes - which can be environmentally harmful in itself - for leaders to get the information they need to develop environmentally friendly policies.

However, it is important to note that our model has a few limitations. Notably, when checking the assumptions of our final model, we found that it did not exhibit independent residuals and constant variance. One important factor that contributed to the lack of independent residuals was the nature of our dataset, which did not contain the specific year each record was sourced from. This prevented us from fully answering the scope of our question and restricted our ability to introduce lagged variables and resolve this assumption failure. Our model also fails the constant variance assumption. This persistent heteroskedasticity could be explained by a lack of important predictors that could influence variance, or that the way in which the data in the dataset was collected had a high inherent variability.

Future steps we can take to improve our model in this regard would be to leverage other datasets that specify a time component and include more meaningful predictor variables.

Appendix

Model Summary for Linear Regression Using Lasso

OLS Model After Lasso Feature Selection:

OLS Regression Results

Dep. Variable:	C02 Emissions(g/km)	R-squared:	0.986
Model:	OLS	Adj. R-squared:	0.986
Method:	Least Squares	F-statistic:	6.434e+04
Date:	Mon, 12 May 2025	Prob (F-statistic):	0.00
Time:	20:14:46	Log-Likelihood:	-21092.
No. Observations:	6282	AIC:	4.228e+04
Df Residuals:	6274	BIC:	4.225e+04
Df Model:	7		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	57.2627	1.942	29.479	0.000	53.455	61.071
Engine Size(L)	0.4545	0.192	2.369	0.018	0.078	0.831
Cylinders	1.3559	0.136	9.987	0.000	1.090	1.622
Fuel Consumption City (L/100 km)	5.6721	1.165	4.868	0.000	3.388	7.956
Fuel Consumption Hwy (L/100 km)	4.8251	0.961	5.018	0.000	2.940	6.710
Fuel Consumption Comb (L/100 km)	8.9792	2.115	4.245	0.000	4.833	13.126
Fuel Consumption Comb (mpg)	-0.8686	0.035	-24.681	0.000	-0.938	-0.800
Fuel Type_E	-106.1344	0.544	-195.230	0.000	-107.200	-105.069

Omnibus:	2606.749	Durbin-Watson:	1.612
Prob(Omnibus):	0.000	Jarque-Bera (JB):	59666.596
Skew:	1.450	Prob(JB):	0.00
Kurtosis:	17.817	Cond. No.	1.01e+03

VIF Values

Before Removing Redundant Variables

1	Engine Size(L)	8.910096
2	Cylinders	8.161044
3	Fuel Consumption City (L/100 km)	2226.387990
4	Fuel Consumption Hwy (L/100 km)	623.630601
5	Fuel Consumption Comb (L/100 km)	5047.427138
6	Fuel Consumption Comb (mpg)	8.446610
7	Fuel Type_E	1.911131

After Removing Redundant Variables

1	Engine Size(L)	8.898367
2	Cylinders	7.630490
3	Fuel Consumption Comb (L/100 km)	4.868929
4	Fuel Type_E	1.644121

Bibliography

1. Dataset:
<https://www.kaggle.com/datasets/debajyotipodder/co2-emission-by-vehicles/discussion/400998>
2. <https://www.canada.ca/en/services/environment/weather/climatechange/climate-action/federal-actions-clean-growth-economy/clean-transportation.html>